

BUSITEMA UNIVERSITY AILAB NLP INTERNSHIP PROGRAM

DAY 1: TOPIC CLASSIFICATION

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1. INTRODUCTION

This task focuses on topic classification, a core Natural Language Processing (NLP) problem where text is categorized into predefined classes. The objective was to train multiple machine learning and deep learning models on a chosen dataset and evaluate their performance comparatively.

2. DATASET

The dataset used for this task consists of textual data stored in the column “**story**” and corresponding category labels stored in “**section**”.

Dataset Characteristics:

- Input: Text (news stories)
- Output: Topic labels (sections)
- Source: Provided dataset (Train and Test CSV files)

Preprocessing Steps:

- Removed missing/null values
- Renamed columns for clarity (story → text, section → label)
- Encoded labels into numerical format using Label Encoding
- Split dataset into training (80%) and validation (20%) sets
- Applied:
 - **TF-IDF vectorization** for classical models
 - **Tokenization** for transformer models (BERT, RoBERTa)

3. MODELS IMPLEMENTED

The following models were trained and evaluated:

a. Classical Machine Learning Models

- Support Vector Machine (SVM): Effective for high-dimensional text data
- Random Forest (RF): Ensemble of decision trees improving robustness
- XGBoost (XGB): Gradient boosting model with strong predictive performance

b. Neural Network

- Feedforward Neural Network (NN):
 - Input: TF-IDF features
 - Architecture: Dense layers with ReLU activation

c. Transformer-Based Models

- BERT (Bidirectional Encoder Representations from Transformers):
Pretrained model fine-tuned for classification
- RoBERTa:
Improved version of BERT with optimized training

d. Ensemble Model

A majority voting ensemble combining predictions from all models was implemented to improve overall performance.

4. EVALUATION METRICS

Models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score

These metrics provide a comprehensive evaluation for multi-class classification tasks.

5. RESULTS

MODEL COMPARISON TABLE		
	MODEL	ACCURACY
0	SVM	0.970511

1	Random Forest	0.933814
2	XGBoost	0.944299
3	Feed forward Neural Network	0.972477
4	BERT	0.982307
5	RoBERTa	0.971822
6	Ensemble	0.973788

6. DISCUSSION

Among all models, **BERT achieved the highest accuracy (98.23%)**, making it the best-performing model for this task. This is because BERT captures deep contextual relationships in text using bidirectional attention, allowing it to understand meaning better than traditional models.

The Feed forward Neural Network, RoBERTa, and SVM also performed strongly, all achieving accuracies above 97%, showing that the dataset is well-structured and separable.

Classical models such as XGBoost and Random Forest performed moderately well but were limited by their inability to capture contextual meaning in text.

The ensemble model achieved high performance (97.38%), but it did not surpass BERT, indicating that BERT alone was already highly effective for this classification task.

7. CONCLUSION

This assignment demonstrates that transformer-based models, particularly **BERT**, are highly effective for text classification tasks.

BERT outperformed all other models due to its ability to capture context and semantic meaning in text, making it the most suitable model for this dataset.